

Prompting using COMET from Perplexity

Perplexity AI is a GenAI platform that works as an AI powered web browser and also generates text when prompted with questions or instructions. We would evaluate the model's response quality and accuracy to various prompts as outlined below -

1. **Instructional** - Here , I prompted the platform to create an MCQ questionnaire with 4 choices , 1 correct.

Provide an MCQ quiz of 3 questions with 4 choices (1 correct answer) to test the depth of knowledge for an AIML student on Linear Regression. Do not provide the answers yet , unless prompted .

Response - Below in the screenshot

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In simple linear regression, you model the relationship between input and output as $y = wx + b$. What is the main purpose of the parameter w in this model?

A It shifts the line up or down to reduce bias.

B It minimizes overfitting by penalizing large weights automatically.

C It represents the prediction error for each training example.

D It controls the steepness of the line, capturing how changes in x affect y .

✓ Correct

This option correctly describes the role of w in simple linear regression. In the model $y = wx + b$, w is the slope, or weight, associated with the input feature x . It tells you how much the predicted output y changes when x increases by one unit, assuming all else stays the same. If $w = 3$, then increasing x by 1 increases the prediction by 3 units; if $w = -2$, increasing x by 1 decreases the prediction by 2 units.

Thinking of w as controlling the steepness of the line helps you interpret the model: a larger magnitude of w means a stronger effect of x on y , and the sign of w indicates whether the relationship is positive or negative. During training, the learning algorithm adjusts w to minimize a loss function, often mean squared error, so that this slope best matches the trend in the training data. This interpretability is one of the reasons linear regression is widely used and taught early in AIML courses.

←

1 of 3

→

Make easier

Make harder

More questions

Customize with a follow-up

2. **Completion** - Few of the impacts of the ongoing conflict in the Middle East on oil prices

Some of the main impacts are:

- **Higher crude oil prices** because conflict raises fears of supply disruption, especially around major shipping routes like the Strait of Hormuz.
- **More volatile fuel costs** as markets react quickly to new attacks, sanctions, or shipping risks.
- **Higher inflation** since more expensive oil feeds into transport, manufacturing, and food prices.
- **Pressure on oil-importing countries** because their trade balances worsen when they have to pay more for energy imports.
- **Possible gains for oil exporters** if they can keep production and exports steady while prices rise.
- **Stronger market uncertainty** which can also affect stocks, currencies, and central bank policy expectations.

In short, Middle East conflict often pushes oil prices up, but the size and duration of the effect depend on whether supply routes or production are actually disrupted.

3. **Example-**

On a **day walking safari** in the wilderness of Kruger National Park, one of the key preparations against adversities is to keep plenty of drinking water. What other things should be prioritized?

Response -

For a **day walking safari** in Kruger, besides plenty of drinking water, the next priorities are sun protection, proper footwear, and insect/malaria precautions. You should also carry a small first-aid kit and wear neutral, comfortable clothing suited to heat and bush terrain.

What to prioritize

- Sunscreen, sunglasses, and a wide-brimmed hat, because the African sun is intense and shade can be limited.
- Closed, comfortable walking shoes or trainers, since open sandals are not suitable for dry riverbeds and thick bush.
- Light, breathable, neutral-colored clothing, which helps with heat and avoids drawing attention.
- Insect repellent and malaria precautions, since Kruger is a malaria area and mosquito protection matters.
- A small medical kit with any personal medication, since supplies may not be easy to get in the park.

Useful extras

- A lightweight rain jacket, because weather can change and summer showers are possible.
- Binoculars if you want better wildlife and bird viewing without getting too close.
- A flashlight for low-light conditions or if the walk finishes near dusk.
- A camera with spare batteries or memory cards, if you want to document the walk.

Safety mindset

Stay close to your guide, keep noise low, and don't carry strong scents like perfume or body spray, since wildlife etiquette and personal safety matter on foot. Avoid heavy, bulky gear so you can move comfortably and stay alert while walking.

Insight analysis on the responses

Analyzing the effectiveness of the responses, it is found that the task-driven Instructional prompt provided a specific **relevant** solution in the form of 3 MCQs. The question captures the fundamental essence of Linear Regression (weights of param), interspersed with incorrect choices, making a candidate ponder over choosing the right answer. The 3 additional unsolicited options (Make easier , harder , More questions) provides a **creative and unique** touch. Furthermore, the structure of the response starting with the options , followed by explanations when a choice is selected , along with further fine tuning the difficulty level makes the response extremely **coherent** and intelligent as it anticipated the difficulty level was never mentioned in the prompt .

For the completion prompt, the agent accurately identified major impacts prioritized point wise.. The beauty of the response lies in identifying current affairs and hence well reasoned attributing the higher price caused due to blockage of the Hormuz.

Furthermore, it also guessed the volatility impact and the aftershocks being felt on markets and oil importing nations. However , the prompt was specifically about impact on oil prices , including inflation and FX into the response might seem an overengineered prompt .

The summary is well crafted making the response **coherent** (as the impacts are prioritized in a top down approach) to an extent but not necessarily **accurate**. In addition, it does include certain **irrelevant** outcomes such as bank policy etc. in an attempt to make itself more **creative**.

Finally, our agent does grasp the challenges encountered in a jungle safari and responds specifically identifying the location as South Africa (from Kruger Park). The contextual comprehension of the unique setting makes the response highly relevant and accurate. There are multitude of priority objects and precautions to be considered and the model groups them well in order of must haves and accessories separately. It identified Africa as a hot place, probably from the in-context learning (*plenty of water*) and also lists appropriate clothing and possibility of injuries (medical kit suggestion). Since the place is a rainforest , suggestion on jacket and comprehensively covers **accurate** requirements and undoubtedly relevant.

The response prioritizes safety of the walker and its assumption that the tour is guided adds a distinct level of **intellectuality** to its response. A deeper analysis shows that it exhaustively covers all challenges with regards to weather (complemented with sunglasses and binoculars on a clear day) , possibility of wet days , appropriate footwear, protection against insects. Intersepersing the points with recreational

necessities such as binoculars, camera etc in an ordered priority list adds both **coherence and creativity** at the same time.